

Design, construction and characterization of a single unit external cavity diode laser coupled tapered amplifier system for atomic physics

Stephen R. Wells^{a,*}, Masabumi Miyabe^b, Shuichi Hasegawa^{a,c}

^a Department of Nuclear Engineering and Management, School of Engineering, University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-8654, Japan

^b Collaborative Laboratories for Advanced Decommissioning Sciences (CLADS), Japan Atomic Energy Agency, Tokai-mura, Naka-gun, Ibaraki 319-1195, Japan

^c Nuclear Professional School, University of Tokyo, 2-22 Shirane, Shirakata, Tokai-mura, Naka-gun, Ibaraki 319-1188, Japan

HIGHLIGHTS

- Construction details for an 'all-in-one' ECDL-TPA laser system.
- System is miniaturizable, portable and built using standard laboratory components.
- Heat related design issues and solutions are fully described.
- Output beam spatial mode profiling of ECDL, TPA and ECDL-TPA is provided.
- Proven application to atomic physics, hyperfine structure of Ca readily resolved.

ARTICLE INFO

Keywords:

External cavity diode laser
Tapered amplifier
Laser
Atomic physics

ABSTRACT

In atomic physics experiments, external cavity diode lasers (ECDLs) are often coupled with tapered amplifiers (TPAs) to provide a highly tunable laser source that also has relatively high output power. Usually these systems are mounted on separate baseplates in separate housings, and when the coupling optics are taken into account, the entire system can take up a large amount of optical table space. Here we report the design, construction, coupling and characterization of an ECDL-TPA system where the ECDL, TPA and coupling optics are mounted on a single, enclosable baseplate. During design and construction, thermal management had a strong influence on the final design, therefore thermal measurements and discussion is emphasized. We report that the output power and frequency was stable over the measurement period. The system is composed of commercially available optical components, and readily machined mounting parts. The system exhibits the possibility of further miniaturization for space-limited applications, and portability.

1. Introduction

Since the first semiconductor laser operation was first demonstrated from GaAs junctions [1–3], laser diodes have become a ubiquitous component in the physics laboratory and in society at large. This is due to their comparatively wide wavelength tunability, narrow linewidth and indeed, cost-effectiveness. In modern atomic physics laboratories, laser diodes are indispensable for providing reliable and easily set up systems for specified wavelengths. Recent applications in atom/ion trapping, for example: atomic clocks [4,5], quantum bit investigation and manipulation [6,7], and fundamental atomic/molecular physics [8,9] all involved the use of laser diodes. In addition, a wide range of fundamental physics may be investigated with the use of these systems, such as gravimetric metrology [10] and physical constant variation or

CP-violation studies [11].

For use in the laboratory, these laser diodes are often put in a construction where the lasing cavity is extended. Some form of frequency selection/control optics returns a single-frequency beam into the diode cavity, effectively locking the diode output to this lasing mode. Therefore, changing the returning beam properties allows the main output wavelength to be controlled, that is, a highly tunable laser system is achievable using relatively simple constructions and components. Such systems are generally called external cavity diode lasers (ECDLs). See e.g. Ref. [12] for a review of typical ECDL forms and properties (N.B. ECDL is designated ECTL in the reference).

For experimentalists, it would seem that the most common practical ECDL designs are the 'Arnold type' [13], or 'Hänsch/Ricci type' [14] in the Littrow or Littman-Metcalf cavity configurations. The Hänsch/Ricci

* Corresponding author.

E-mail address: wells@lyman.q.t.u-tokyo.ac.jp (S.R. Wells).

<https://doi.org/10.1016/j.optlastec.2020.106118>

Received 15 August 2019; Received in revised form 11 December 2019; Accepted 31 January 2020

Available online 14 February 2020

0030-3992/ © 2020 Elsevier Ltd. All rights reserved.

type locates the diffraction grating and grating positioning screws separate to the diode mount-piece, whereas the Arnold type fixes the grating mount arm to the laser diode mount. There are also stability focused monolithic mount designs adapting some form of these types [15], but new designs are continually being produced. For example, the use of micro-electro-mechanical systems (MEMS) is reported to allow for improved tuning capability while also enabling high miniaturization [16].

While this tunability is a highly desired property, often the main drawback is the relatively low useful output laser powers, generally on the order of 1–10 mW. A common method of obtaining higher power while maintaining tunability, is to use an ECDL output as a seed beam for an additional gain medium, where this gain medium is often the active region of a tapered amplifier (TPA) [17]. TPAs are semiconductor devices similar to laser diodes, but in which the gain medium has geometry where the ‘input’ facet medium is narrow, broadening over the length of the chip to a wider output facet, hence ‘tapered’. The tapering allows the optical power density to be spread as the propagation and amplification of the input beam progresses, with the large surface area reducing heat production and enabling large current injection for high amplification. Generally, TPAs are used with an ECDL seed beam, and since the amplification preserves the seed beam characteristics, a highly tunable, high power laser output may be realized.

In practice, the TPA is mounted on either commercially available mounting blocks, or on homemade blocks [18], in a separate enclosure to the ECDL. Here we investigate siting the ECDL and TPA components on a single baseplate, in a single enclosure, and characterize the system accordingly. We aim for this to lead to possible further miniaturization of the ECDL-TPA system for applications where portability is required, for example: remote analysis of dangerous substances, or more generally, transportable experimentation capability. In addition, the characterization seeks to provide confirmation that the ECDL-TPA system maintains the stability, tunability and high output power of more usual, separate ECDL and TPA experimental setups.

2. Material and methods

2.1. Design and construction of the ECDL-TPA system

We were strongly motivated to design a compact ECDL-TPA system as optical table space was extremely limited. Additionally, frequent collaboration with an external laboratory some distance away meant that portability, or at least, uncomplicated preparation for transport was also a requirement. Consequently, the decision was made to site both ECDL and TPA components, along with the intermediary coupling optics on a single enclosable aluminum baseplate. The schematic diagram for this baseplate, outlining dimensions and component locations is provided in Fig. 1.

The baseplate dimensions were specifically chosen to enable interface with a commercially available polycarbonate enclosure. That is, with laser components in place, the enclosure may be placed over the components, closely fitting with the baseplate, to provide effective laboratory environment isolation (necessary for stable operation of the ECDL-TPA) and/or convenient packaging for system transport. The outer enclosure dimensions are 380 × 280 mm, providing a balance between small optical table footprint and enough space for easy placement and adjustment of the internal optical components.

The two M6 holes in the central portion of the baseplate allow direct fixing of the plate to standard optical tables, stabilizing the system on the table and ensuring good thermal contact with the table as ultimate heat sink.

The ECDL consists of a standard Hänsch-type external cavity construction [14] operating in the Littrow configuration (first-order diffraction returned to the diode), but with an added ‘on-board’ mirror to direct the laser output in a direction parallel to the long-edge of the ECDL base, similar to the adaptation of Ref. [19]. This generally enables

a more straightforward base positioning procedure. The laser diode was a Thorlabs L852P100 model, this 852 nm wavelength chosen for calcium spectroscopy experiments. Maximum intra-cavity power ~100 mW, maximum extra-cavity power ~20 mW. The fast axis divergence was arrested and the laser beam collimated by an aspheric lens (Thorlabs A390-B, $f = 4.60$ mm, $NA = 0.53$), and the external cavity formed in the Littrow configuration by a Thorlabs GR13-1206 grating, blaze wavelength 750 nm and groove density 1200 grooves per mm. The temperature was stabilized by a Peltier element operated by Thorlabs TED200C controller with input from an AD590 transducer mounted in the ECDL base.

The mount for the TPA followed the design of Kangara et al. [18]: two copper blocks support custom-made rough- and fine-threaded aluminum cylinders housing aspheric collimation lenses Thorlabs C230TMD-B ($f = 4.51$ mm, $NA = 0.55$), as well as the c-mount tapered amplifier itself. The lenses are placed either side of the TPA; the lens at the output collimates the highly divergent fast axis, while the lens at the input focusses the ECDL seed beam into the small TPA input facet for amplification. While the original design functions well in the output direction, we found replacing the collimation cylinder on the input side with the aspheric lens mounted on an xy-adjustable lens holder, itself mounted on a z-axis stage, provided an easier and perhaps more accurate method of beam alignment for focusing the seed beam into the ~3 μ m TPA input aperture. We found that small machining inaccuracies caused the collimation cylinder to be loose or otherwise off-axis in the threaded hole, preventing optimal coupling. This problem is negligible in the output direction. In our setup, the TPA was an Eagleyard EYP-TPA-0850-01000-2006-CMT04-0000 model, center wavelength 850 nm, maximum output power ~1 W, however since sustained continuous wave operation was required for our experimental purposes and therefore for the prevention of high-power related optical damage to some of our optical components, this was generally limited to ~250 mW for experimentation. Continual operation at this output power produces a large amount of heat deposited into the copper mounts. This is moderated by an additional TED200C controlled Peltier element and AD590 sensor. The heat is ultimately transferred into the heat sink system described as follows: on top of the baseplate, a single aluminum block (220 × 100 × 34 mm) is thermally fixed. This block acts as the main heat sink for both the ECDL and TPA mounts, which are affixed to this block. During the initial design process, separate heat sink blocks were used for the ECDL and TPA mounts respectively. However, the large amount of heat produced during normal operation of the TPA was not able to be adequately removed by the relatively small heat sink. The thermal expansion of the heat sink block moved the TPA input facet out of the focus of the seed beam, quickly reducing the coupling between ECDL and TPA to zero. This was readily observed by measuring the power output of the TPA with respect to time, as recorded in Fig. 2 for both the power output with the individual heat sinks and the single heat sink cases. We would also assume that coupling condition may be improved under the single heat sink case since any thermal expansion that does occur, would cause both the TPA input aperture and the seed beam to rise in tandem, local variations in expansion notwithstanding.

The individual heat sink case shows rapid decoupling of the seed beam, as indicated by the 50% reduction in output power over a short 10-minute window. Replacement with the single heat sink produced no change in output power condition over the 3-hour measurement period and is assumed to allow effectively stable constant operation.

2.2. Methods in seed beam coupling

When current is applied through the TPA bandgap in the absence of a seed beam, amplified spontaneous emission (ASE) is produced and emerges from both facets of the amplifier. Even without laser amplification and emission, the ASE can be reasonably strong – we measured the output ASE under medium current application (950 mA) to be 10 mW, while the input facet ASE was at least 1–2 mW. During

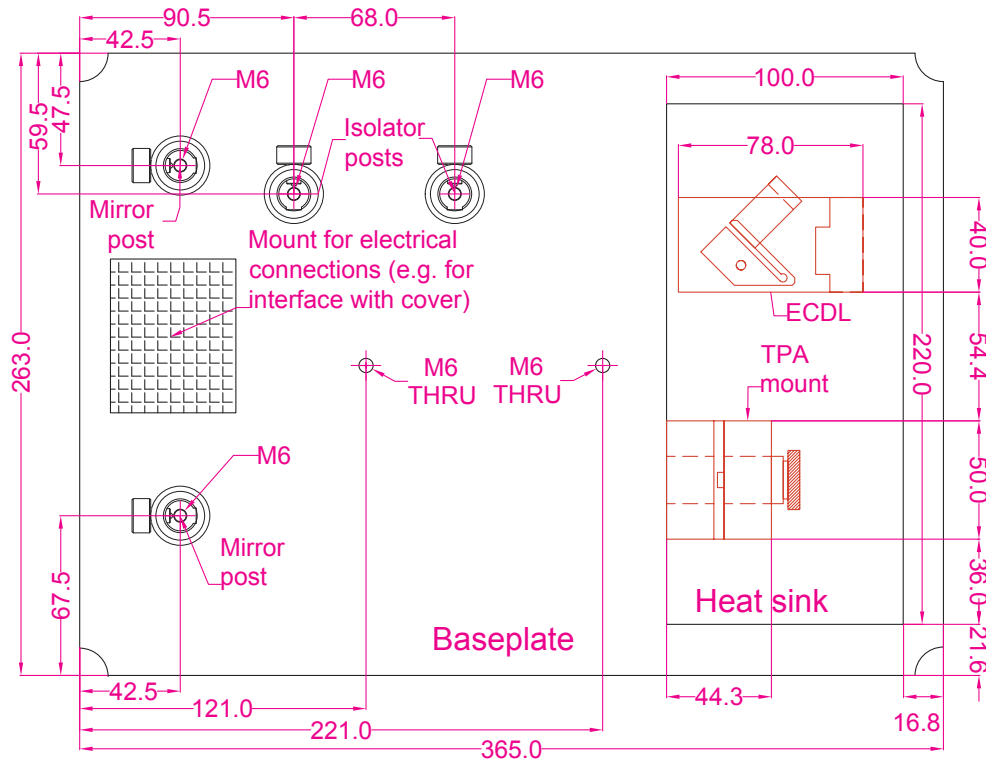


Fig. 1. [2 column image]: Plan-view schematic diagram of the ECDL-TPA baseplate. Baseplate thickness 10 mm. All measurements in mm.

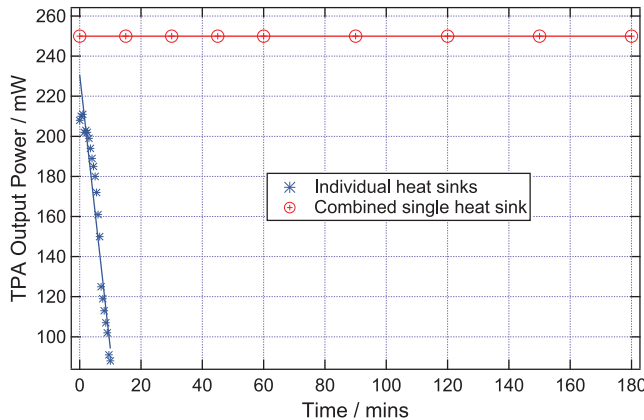


Fig. 2. [1 column figure]: Power stability of the ECDL-TPA system for the case of individual heat sinks, and the combined single heat sink. Fit lines are linear.

coupling, this ASE becomes aligned with the ECDL seed beam so it is vitally important to ensure appropriate isolation of the ASE to prevent it entering the ECDL cavity and causing significant instability. During construction, we inserted a single 30 dB isolator into the seed beam path to prevent back-reflection from the seed beam itself as well as for extinguishing the TPA ASE, and coupled the ECDL and TPA. However, instability in the output mode profile was observed. A stable mode profile was obtained after insertion of a second 30 dB isolator into the beam path, for total ~60 dB isolation.

For coupling, we found the following method most optimal. With the ECDL Littrow configuration optimized and output collimated, passing through the optical isolators, initial coupling may be achieved by overlapping the ECDL beam with the collimated ASE emitted from the TPA as it counter-propagates. Space to accomplish this is limited, but we found the space between the TPA input facet collimation lens and the preceding mirror is sufficient. With reference to Fig. 1, this is the space on the opposite side of the baseplate to the optical isolator

section. During the coupling, it is important to note that the TPA is a very delicate device. Care must be taken to ensure that the seed beam does not damage the device during coupling – directing the ECDL beam at the TPA without running current through the TPA is reportedly especially dangerous. We found the 950 mA supply current producing the 1–2 mW of rear facet ASE to be sufficient for the alignment procedure, though these values are arbitrary. Two irises placed in the beam path may be used to aid the overlap process, or simple placement of a thin (IR) viewing card may be enough to observe both beams simultaneously for superimposition.

Once initial coupling is confirmed via power meter measurement of the TPA output – a noticeable increase of output power will be observed even for small coupling efficiencies – then optimization may be performed. This is mainly done by iterative alignment (‘walking the beam’) via the mirror pair, and then perfected with delicate adjustments to the rear collimation lens position, repeating the process until output power is maximized. It is important to note that the coupling is highly dependent on the seed beam polarization orientation with respect to the plane of the TPA active gain region. If the seed beam polarization orientation is not optimally aligned after exiting the final Faraday isolator, a half-waveplate may be inserted in the beam to rotate the orientation, with output power maximization corresponding to the optimization of the orientation.

As noted above, the TPA output fast axis is collimated using the aspheric lens positioned in front of the output facet. The slow axis divergence may also be relatively large (the divergence in the amplifier listed above is 15° for example), and will also require additional collimating optics. For our system, a cylindrical lens (Thorlabs LJ1567L1-B, $f = 100$ mm) was placed external to the enclosure for the slow axis collimation.

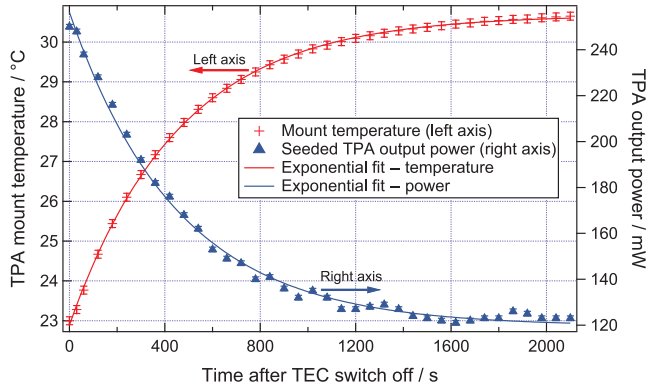


Fig. 3. [1 column figure]: TPA mount temperature and TPA output power versus time after the TPA mount thermal stabilization system was switched off. Fitting lines are exponential in character. Error bars represent experimental uncertainties.

3. Results and discussion

3.1. ECDL-TPA thermal characterization

Characterization of the ECDL-TPA output enables confirmation of the system capability and stability for atomic physics experiments. Firstly, the output amplification was corroborated with manufacturer specifications, and the seed beam coupling confirmed stable over at least a 3-hour period (see Fig. 2).

As a further display of the importance and capability of the heat sink system with the combined heat sink, an experiment was undertaken wherein the ECDL seed beam was coupled to the TPA and the applied currents set to achieve power output at 250 mW, corresponding to the experimental operating power as described in Section 2. Once stable operation was achieved, the thermal stabilization of the TPA was turned off, and the power output and mount temperature recorded over time, with the ECDL and TPA drive currents remaining the same. Fig. 3 displays the results of this. One of the most striking points is that the output power begins to decrease almost instantaneously, with a ~ 20 mW drop occurring in the first 30 s after the stabilization shut-off. The

exponential characteristic of the temperature increase corresponds to the exponential function of the transient solution to the inhomogeneous heat diffusion Eq. (1):

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T + q \quad (1)$$

with T being temperature, t being time, κ being thermal diffusivity and q being heat flux density, a source term accounting for the in-operation TPA.

Making the assumption that the temperature difference from start to finish $\Delta T \sim 7.5$ °C is a small change, we can assume the change in mount volume is related by the standard thermal expansion Eq. (2):

$$\frac{\Delta V}{V} = \alpha \Delta T \quad (2)$$

where V is volume and α is the coefficient of thermal expansion.

As an exercise, if we take the linear expansion form of Eq. (2): $\frac{\Delta L}{L} = \alpha_L \Delta T$, and with the linear expansion coefficient of copper being $16.5 \times 10^{-6} \text{ m/m}\cdot\text{K}$, for the overall temperature change of Fig. 3 $\Delta T = 7.65\text{K}$, the linear fractional change in one dimension of the copper mount is $1.3 \times 10^{-4} \text{ m/m}$. The height of the TPA waveguide in the copper mount is 11.6 mm, and so the actual length change is $\Delta L \approx 1.5\mu\text{m}$. The input aperture of the TPA is $3 \mu\text{m}$ [20], and the (Gaussian) waist size at the focal point of the TPA input lens is $1 \mu\text{m}$. Therefore this length change would cause the input aperture to ‘shift’ away from the seed beam optimal coupling location, and effecting the drop in output power observed in Fig. 3. MATLAB [21] simulations solving Eq. (1) allow the heat flow through the mount to be visualized. Using these solutions at time $t = 2100 \text{ s}$ (see Fig. 3), heat maps through the copper mount holding the TPA are displayed in Fig. 4. While the bulk areas of the copper reveal small temperature gradient, the relatively thin ($\sim 4.3 \text{ mm}$) region supporting the TPA c-mount structure shows a more stratified heat diffusion pattern. This localization of the heat flow gives rise to the strong thermal expansion calculated above. Since the TPA requires closely positioned collimating lenses either side of the waveguide path, it is unfeasible to increase the mounting block bulk to compensate. Consequently, the thermal control provided by the TEC system is more than justified.

This exploration of the thermal conditions and effects arising from the use of the TPA in this system configuration is intended to emphasize

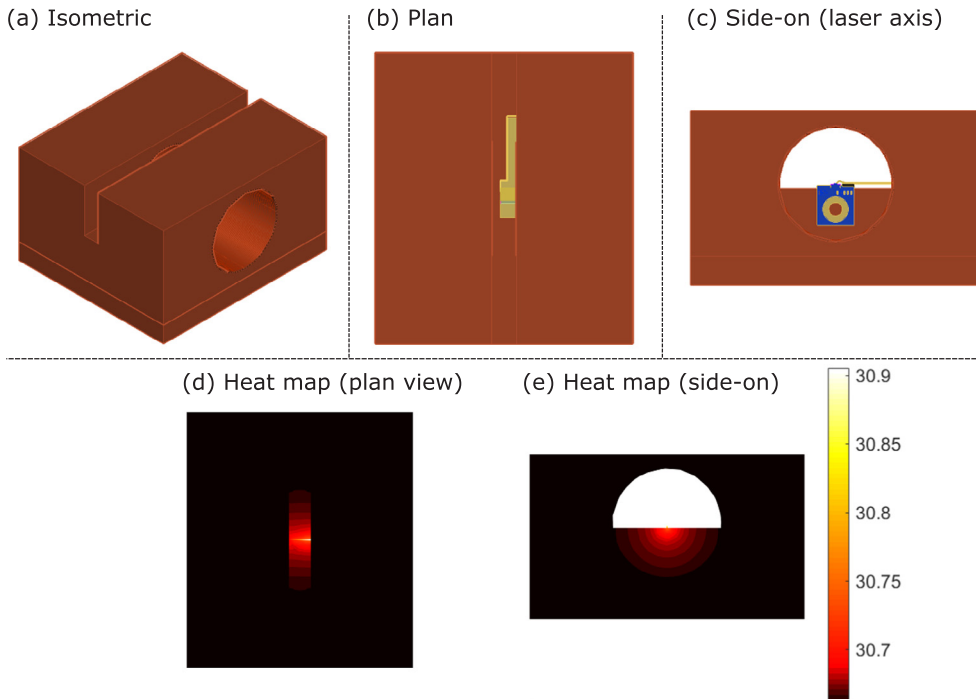


Fig. 4. [1.5 column figure]: Diagrams of the TPA copper mount block and MATLAB simulated heat maps of the block for the 2100 s (final) time point as corresponding to Fig. 3. Map (a) is an isometric view, showing the threaded hole for the collimating lens mount; (b) is the view from above, the yellow colored object being the TPA package and electrical connector; (c) is the side-on view highlighting the TPA package positioning, with the TPA waveguide region at the axis center of the collimation lens; (d) and (e) are heat maps at the final time point where the heat flow has reached steady state. Note the bright white strip representing the TPA waveguide area in (d).

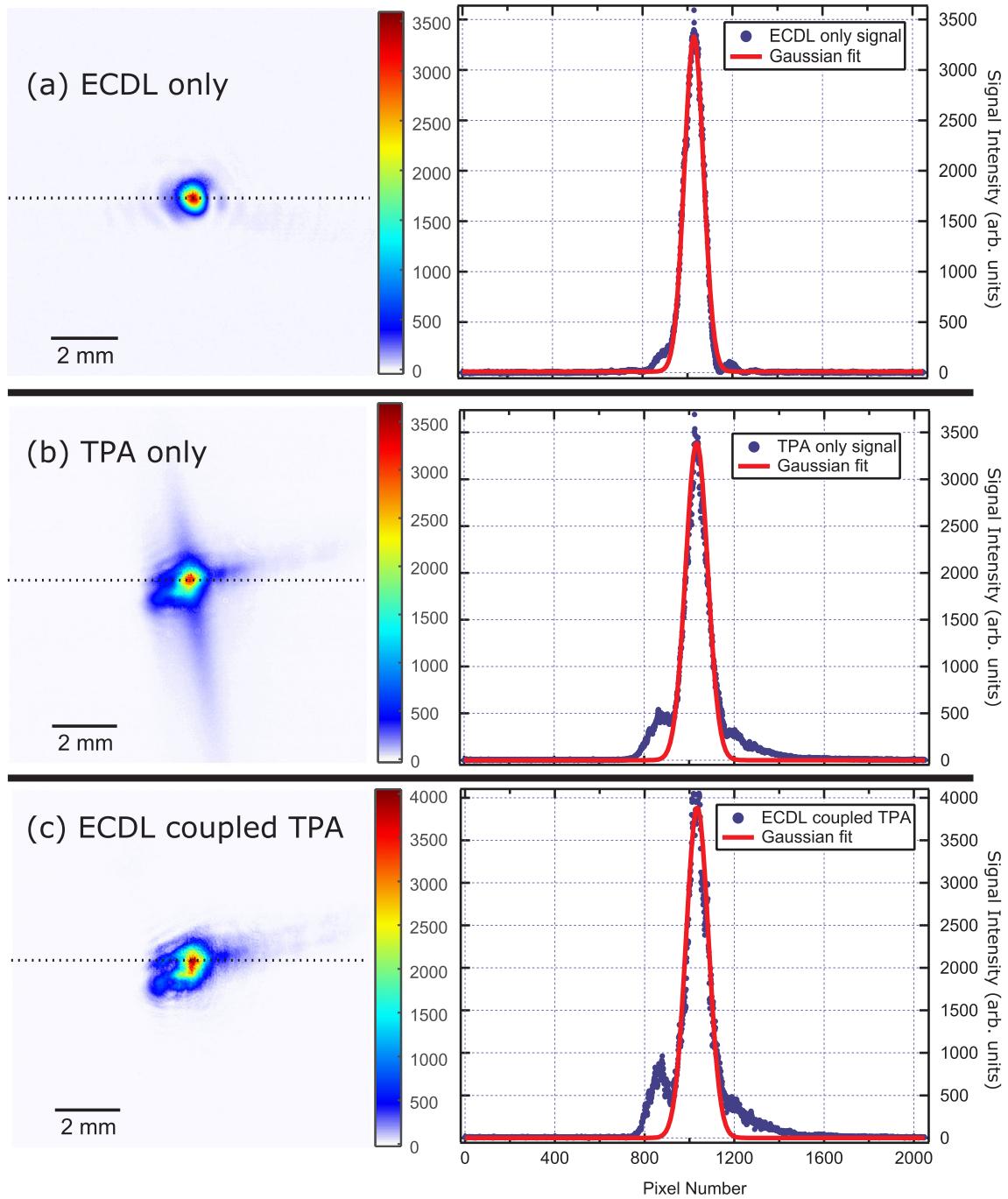


Fig. 5. [2-column figure]: Beam spatial profile images and associated 1D plots of the row of pixel data intersecting the spot of highest intensity, of (a) the ECDL laser output; (b) the TPA output and (c) the coupled ECDL-TPA laser beam output. 1D plots are taken from the pixel row data indicated by the dotted line on the beam spot profiles. A Gaussian fit is provided in the plots to enable qualitative comparison.

that good control of the temperature is extremely necessary for stable TPA output, as well as ECDL mode conditioning.

3.2. ECDL-TPA output beam spatial profile

TPA devices are well known for having non-Gaussian output beam profiles, which in some cases can cause difficulty in experiments requiring a clean single-mode profile. We therefore investigated the beam profile of the ECDL, TPA and coupled ECDL-TPA output beams as part of the system characterization. Fig. 5 displays these beam profiles as obtained using a Gentec Beamage-4 M laser beam profiler, alongside Gaussian fit plots of the 1D profile taken through the center of the laser

spot.

The ECDL only beam in Fig. 5(a) appears to have a reasonably clean Gaussian profile, as recorded by the beam profiler. Some small 'wings' may be observed on the 1D plot however. The TPA amplified spontaneous emission (ASE) emitted by the TPA without the ECDL beam coupled shows significant deviation from a Gaussian profile, as should be expected from the multimode constituent of this beam (Fig. 5(b)), and indeed the 'wings' in the 1D plot are large. To obtain Fig. 5(c), neutral density filters were placed in the beam path to attenuate the high intensity ECDL-TPA beam for capture by the beam profiler. Again significant deviation from a standard Gaussian profile is observed for the coupled ECDL-TPA case, with the shape appearing to be an

‘amplified’ version of the non-coupled case in Fig. 5(b), indicating that the spatial mode output is dominated by the TPA. In the 1D plot, the ‘wings’ appear to also be amplified to as much as 25% of the intensity of the main spot. While the profile is not ‘clean’, it should be noted that it was still able to be coupled into a single mode optical fiber for wavemeter insertion and also achieve the experimental objective of the ECDL-TPA system, as discussed in the following section.

3.3. ECDL-TPA system capability – Atomic spectroscopy

In terms of experimental capability, this ECDL-TPA system was constructed in order to provide a laser of sufficient intensity to excite the low-oscillator-strength third transition into the Rydberg states in a three-step resonance ionization transition scheme. The scheme uses the following levels: ground state $4s^2 1S_0 \rightarrow 4s$ $4p^1 P_1^0 \xrightarrow{732.6nm} 4s4d^1 D_2 \xrightarrow{\sim 840nm} 4sn p^1 P_1^0$ Rydberg states. The first and second step transition lasers are provided by homemade ECDLs. All beams are sampled by beamsplitters for wavemeter input for wavelength checking, as well as input into a scanning Fabry-Perot etalon for digital frequency offset control (DFOC) [22] laser frequency stabilization (against a stabilized He-Ne reference laser). The main portion of each laser is arranged to cross perpendicularly with a beam of atomic calcium, as produced by resistive heating in a graphite furnace, housed in a vacuum chamber. To limit Doppler broadening, the higher photon energy first step laser beam (423 nm) is arranged to be collinear but counterpropagating against the second and third step beams at the crossover with the atomic beam. Calcium ions that are produced are guided with ion optics to a quadrupole mass spectrometer for ion identification and quantification. This experimental setup is represented as a schematic in Fig. 6.

In regards to our experimental objective of investigating the isotope shifts of high-lying Rydberg states of calcium, here we provide an example of the suitability and capability of the ECDL-TPA system in achieving that purpose. Applying some additional feedforward electronics to the already highly frequency tunable ECDL component allows for continuous frequency scanning of the high-power ECDL-TPA laser. In laboratory tests, we were able to achieve up to 30 GHz continuous scan range. For the calcium experiments, 3 GHz was sufficient to obtain ion signals from each calcium ion of interest in a continuous scan. Fig. 7(a) shows a composite of the calcium ion signals obtained from such a scan for each isotope, using the transition from $4s4d^1 D_2 \rightarrow 4s30p^1 P_1^0$ where this $n = 30P$ level has been unreported in literature previously. To obtain each isotope shift value, the quadrupole mass spectrometer (QMS) is set in near-simultaneous scan mode for Ca-40 and an additional isotope of interest. The first and second step

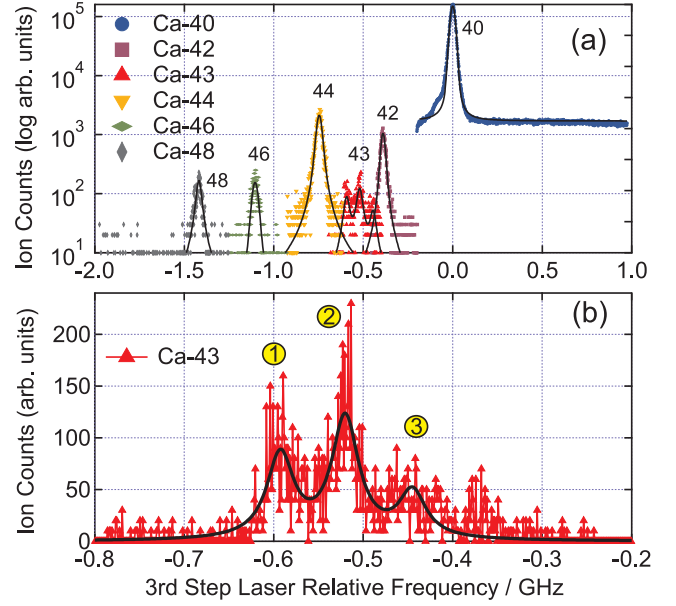


Fig. 7. [1-column figure]: Calcium ion signal as a function of the 3rd step laser (ECDL-TPA) frequency for the transition into the $n = 30P$ Rydberg state. Frequency units are set relative to the majority abundance Ca-40 ion peak location. Solid lines are Voigt function fits to the experimental data. Part (a) displays the full $\Delta\nu = 3$ GHz scan, start location + 1.0 GHz relative to the Ca-40 peak, then decreasing in frequency until stop point -2.0 GHz. All isotopes of interest are observable within this scan range, with Ca-43 and Ca-46 content enriched for easier observation. Part (b) is a closeup of the Ca-43 hyperfine structure, readily resolved with the ECDL-TPA laser system. The three peaks marked 1, 2 and 3 correspond to the three hyperfine levels accessible with this transition: $F = \frac{5}{2}, \frac{7}{2}$ and $\frac{9}{2}$ respectively. The hyperfine peaks are fitted with a Lorentzian fit function (solid line).

transition frequencies are first optimized for Ca-40 excitation (the Ca-40 signal region with magnitude 10^3 from 0.0 to +1.0 GHz corresponds to non-resonant excitation from an additional absorption of a first or second step photon). The ECDL-TPA is then set to scan in frequency continuously over the 3 GHz range. After the Ca-40 peak is confirmed, the first and second step laser frequencies are quickly switched to the transition frequencies corresponding to the isotope of interest, hence the Ca-40 signal drop-off around -0.2 GHz. The other Ca isotope peaks then appear as the third step laser frequency passes through their resonant transition frequencies. In this way, the transition isotope shift values relative to Ca-40 for this Rydberg level may be

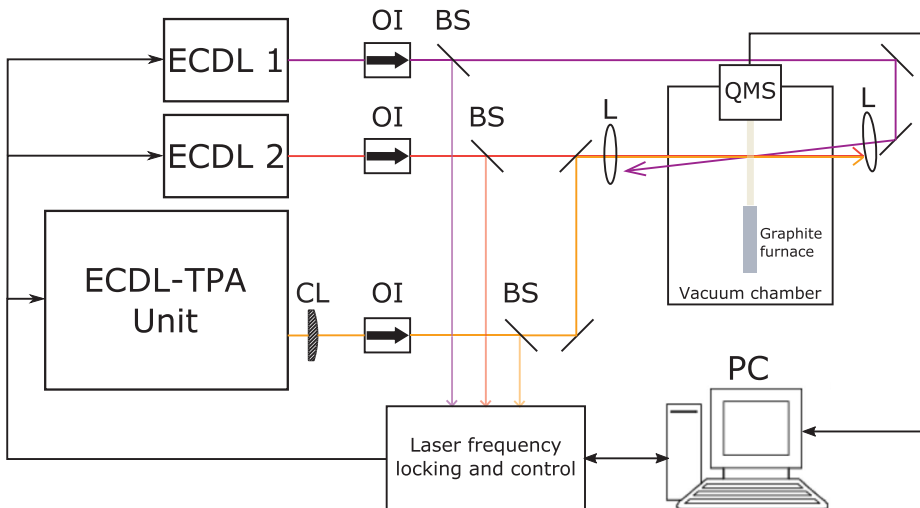


Fig. 6. [1.5-column figure]: Experimental setup for the three-step resonance ionization of calcium. ECDL 1 provides the 422.7 nm beam; ECDL 2 provides the 732.6 nm beam. The ECDL-TPA unit is as described in the text. CL is the cylindrical lens used as a final collimator for the seeded TPA beam's slow axis (see text). OI – optical isolators; BS – beam splitters; L – focussing lenses; QMS – quadrupole mass spectrometer and PC – desktop computer for laser frequency control and QMS control and measurement.

calculated as a simple difference between the Ca-40 peak location and the isotope of interest peak frequency location.

As a more explicit example of the capability of the ECDL-TPA system, Fig. 7(b) shows the frequency resolved hyperfine structure of Ca-43. Ca-43 has a nuclear magnetic moment $I = \frac{7}{2}$, and the transition parameters are chosen such that $J = 2$, $J' = 1$ and $F = \frac{7}{2}$ so that $F' = \frac{5}{2}$. $\frac{7}{2}$ or $\frac{9}{2}$ according to selection rules, in order to reveal the hyperfine structure. Unfortunately, the signal magnitude in Fig. 7(b) is small, and the signal to noise ratio is poor, but the expected peak intensity ratios are roughly correct. In particular, each hyperfine peak with linewidth estimated from the Lorentzian fit function to be 37 ± 4 MHz, is clearly resolved by the scanning ECDL-TPA which implies a laser linewidth much less than the small hyperfine splitting components. Indeed, from this splitting, the hyperfine A coefficient for this level can be estimated (assuming the quadrupole B coefficient ≈ 0) from the interval equation $E_F - E_{F'} = AF$, where E_F and $E_{F'}$ are the energies of the hyperfine levels denoted F and F' . An estimate value obtained from an average of the two individual A values that can be calculated from the three peaks is $A = 18.6$ MHz. This value is a few MHz higher than the trend of A values reported in Ref. [23] indicating that either the trend is non-linear at higher values of n ; the hyperfine B coefficient is non-zero at higher values of n ; or the low signal magnitude is preventing the Voigt fitting function from calculating the peak locations accurately. We note that a sample containing a larger amount of Ca-43 would give a higher signal intensity and possibly a different hyperfine coefficient, and this is unrelated to the laser capability.

Such an experiment confirms several important characteristics of the ECDL-TPA system:

1. Inspection of the Fabry-Perot interferometer signal would indicate the coupling between ECDL and TPA does not result in significant linewidth broadening. The hyperfine peak discrimination of Fig. 7(b) indicates a maximal linewidth of ~ 10 MHz, in keeping with the generally narrow linewidth output of ECDLs.
2. The frequency control of the ECDL allows for consistent frequency control of the coupled output; change of the seed frequency does not significantly alter the TPA output power conditions, at least for standard spectroscopic investigations.

Both of these characteristics indicate the appropriateness of this ECDL-TPA system design for high-sensitivity atomic spectroscopy experiments. The system may also be appropriate for any experimental situation where frequency tunable but relatively high power laser light is required in a compact and portable unit, such as physical research in the field or in space, among other situations.

4. Conclusions

We have introduced an ECDL-TPA laser system for atomic physics and other bench-top experimental objectives requiring laser wavelength tunability with relatively high power. The design places the ECDL, TPA and associated coupling optics on a single baseplate, which enables system portability and optical table space conservation, while providing a stable and reliable amplified output. As the system is homemade from a mixture of standard laser diode mount designs and readily commercially available optics, the component siting and the components themselves may be freely changed depending on individual requirements. In addition, since the size of the system was somewhat arbitrarily chosen based on available commercial enclosures, further miniaturization of the system is possible, should appropriate heat management also be possible. As a result of investigating the thermal parameters of the system, we noted that excellent control of the heat sinking and heat management components is absolutely necessary for stable operation of the ECDL-TPA unit, and so care must be taken should any adjustments to the design that impact the heat flow be

undertaken. We measured the system performance for both optimal and sub-optimal thermal conditions to analyze the parameters involved, as well as to emphasize the effect on system performance. Further characterization of the output beam spatial mode profile was investigated by beam profiler camera, and while the shape is imperfect, we confirmed it may still be coupled into single mode fiber, as well as be used for the precision experiments it was designed for.

Finally and practically, we successfully carried out atomic spectroscopy experiments using this system and were motivated to provide a full description for other laser experimentalists.

CRedit authorship contribution statement

Stephen R. Wells: Writing - original draft, Writing - review & editing, Methodology, Investigation, Formal analysis. **Masabumi Miyabe:** Methodology, Investigation, Resources, Writing - review & editing. **Shuichi Hasegawa:** Conceptualization, Methodology, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to gratefully acknowledge Yuta Yamamoto and Masaaki Kato for technical assistance in the laboratory.

S.R.W acknowledges personal support in the form of a scholarship from the Japan Ministry of Education, Culture, Sports, Science and Technology.

Funding

This work was supported by JSPS KAKENHI [grant number 15H04240].

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.optlastec.2020.106118>.

References

- [1] R.N. Hall, G.E. Fenner, J.D. Kingsley, T.J. Soltys, R.O. Carlson, Coherent light emission from GaAs junctions, *Phys. Rev. Lett.* 9 (1962) 366–368, <https://doi.org/10.1103/PhysRevLett.9.366>.
- [2] M.I. Nathan, W.P. Dumke, G. Burns, F.H. Dill, G. Lasher, Stimulated emission of radiation from GaAs p-n junctions, *Appl. Phys. Lett.* 1 (3) (1962) 62–64, <https://doi.org/10.1063/1.1777371>.
- [3] T.M. Quist, R.H. Rediker, R.J. Keyes, W.E. Krag, B. Lax, A.L. McWhorter, et al., semiconductor maser of GaAs, *Appl. Phys. Lett.* 1 (4) (1962) 91–92, <https://doi.org/10.1063/1.1753710>.
- [4] O. Kock, W. He, D. Świerad, L. Smith, J. Hughes, K. Bongs, et al., Laser controlled atom source for optical clocks, *Sci. Rep.* 6 (1) (2016) 37321, <https://doi.org/10.1038/srep37321>.
- [5] P. Yun, F. Trialet, C.E. Calosso, S. Micalizio, B. François, R. Boudot, et al., High-performance coherent population trapping clock with polarization modulation, *Opt. Rev.* 7 (1) (2017), <https://doi.org/10.1103/PhysRevApplied.7.014018>.
- [6] N. Akerman, N. Navon, S. Kotler, Y. Glickman, R. Ozeri, Universal gate-set for trapped-ion qubits using a narrow linewidth diode laser, *New J. Phys.* 17 (11) (2015), <https://doi.org/10.1088/1367-2630/17/11/113060>.
- [7] H. Levine, A. Keesling, A. Omran, H. Bernien, S. Schwartz, A.S. Zibrov, et al., High-fidelity control and entanglement of rydberg-atom qubits, *Phys. Rev. Lett.* 121 (12) (2018), <https://doi.org/10.1103/PhysRevLett.121.123603>.
- [8] M. Juan, C. Bradac, B. Besga, M. Johnsson, G. Brennen, G. Molina-Teriza, et al., Cooperatively enhanced dipole forces from artificial atoms in trapped nanodiamonds, *Nat. Phys.* 13 (3) (2016) 241–245, <https://doi.org/10.1038/nphys3940>.
- [9] X. Wu, F. Zi, J. Dudley, R.J. Bilotta, P. Canoza, H. Müller, Multiaxis atom interferometry with a single-diode laser and a pyramidal magneto-optical trap, *OPTICA*

- 4 (12) (2017) 1545–1551, <https://doi.org/10.1364/OPTICA.4.001545>.
- [10] Q. Wang, Z. Wang, Z. Fu, W. Liu, Q. Lin, A compact laser system for the cold atom gravimeter, *Opt. Commun.* 358 (2016) 82–87, <https://doi.org/10.1016/j.optcom.2015.09.001>.
- [11] Q. Liu, H. Zou, X. He, G. Chen, Y. Shen, J. Yuan, Trapped Hg⁺ ion crystal generated by isotope-selective photoionization, 013107, *Rev. Sci. Instrum.* 90 (1) (2019), <https://doi.org/10.1063/1.5068692>.
- [12] B. Mroziwicz, External cavity wavelength tunable semiconductor lasers - a review, *Opto-Electron. Rev.* 16 (4) (2008) 347, <https://doi.org/10.2478/s11772-008-0045-9>.
- [13] A.S. Arnold, J.S. Wilson, M.G. Boshier, A simple extended-cavity diode laser, *Rev. Sci. Instrum.* 69 (3) (1998) 1236–1239, <https://doi.org/10.1063/1.1148756>.
- [14] L. Ricci, M. Weidemüller, T. Esslinger, A. Hemmerich, C. Zimmermann, V. Vuletic, et al., A compact grating-stabilized diode laser system for atomic physics, *Opt. Commun.* 117 (5) (1995) 541–549, [https://doi.org/10.1016/0030-4018\(95\)00146-Y](https://doi.org/10.1016/0030-4018(95)00146-Y).
- [15] E.C. Cook, P.J. Martin, T.L. Brown-Heft, J.C. Garman, D.A. Steck, High passive-stability diode-laser design for use in atomic-physics experiments, 043101, *Rev. Sci. Instrum.* 83 (2012), <https://doi.org/10.1063/1.3698003>.
- [16] M. Hoppe, A. Jimenez, H. Rohling, S. Schmidtman, J. Grahmann, H. Tatenguem, et al., Construction and characterization of external cavity diode lasers based on a micro-electro-mechanical system device, *IEEE J. Sel. Top. Quant.* 25 (6) (2019) 2700109, <https://doi.org/10.1109/JSTQE.2019.2912059>.
- [17] G. Bendelli, K. Komori, S. Arai, Y. Suematsu, A new structure for high-power TW-SLA (Travelling wave semiconductor laser amplifier), *IEEE Photonic Tech L* 3 (1) (1991) 42–44, <https://doi.org/10.1109/68.68042>.
- [18] J.C.B. Kangara, A.J. Hachtel, M.C. Gillette, J.T. Barkeloo, E.R. Clements, S. Bali, et al., Design and construction of cost-effective tapered amplifier systems for laser cooling and trapping experiments, *Am. J. Phys.* 82 (8) (2014) 805–817, <https://doi.org/10.1119/1.4867376>.
- [19] C.J. Hawthorn, K.P. Weber, R.E. Scholten, Littrow configuration tunable external cavity diode laser with fixed direction output beam, *Rev. Sci. Instrum.* 72 (12) (2001) 4477–4479, <https://doi.org/10.1063/1.1419217>.
- [20] Eagleyard EYP-TPA-0850-01000-4006-CMT04-0000 specification sheet.
- [21] MATLAB and Partial Differentiation Toolbox Release 2019a The MathWorks Inc. Natick, Massachusetts, United States.
- [22] M. Miyabe, M. Oba, M. Kato, I. Wakaida, K. Watanabe, Development of RIMS apparatus for isotope analysis of calcium in nuclear waste materials, *J. Nucl. Sci. Technol.* 43 (4) (2006) 305–310, <https://doi.org/10.1080/18811248.2006.9711094>.
- [23] P. Muller, B.A. Bushaw, W. Nortershauser, K. Wendt, Isotope shifts and hyperfine structure in calcium 4snp 1P1 and 4snf F Rydberg states, *Eur. Phys. J. D* 12 (2000) 33–44, [https://doi.org/10.1016/S0584-8547\(98\)00108-6](https://doi.org/10.1016/S0584-8547(98)00108-6).